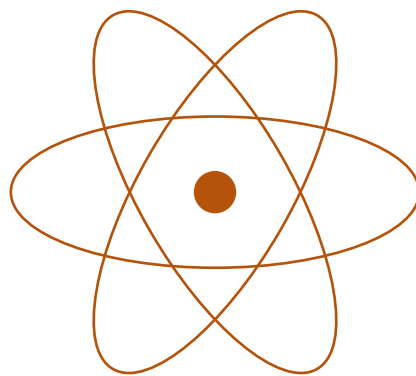


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Units Conversion and Dimensional Analysis

Olympiad Problem Set & Reference

Olympiad Training • Physics Fundamentals

SPhOH Team

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About This Problem Set

This problem set is part of the foundational training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It contains 29 structured problems arranged strictly by increasing difficulty in units conversion and dimensional analysis.

Difficulty Levels

- **Level I** — Fundamental Olympiad problem-solving practice.
- **Level II** — Problems requiring combinations of ideas.
- **Level III** — More serious problems requiring deeper analysis.

Find more at: <https://devfhd.github.io/SPhOH/>.

Units Conversion

Topic Note: The Chain-Link Method

Unit conversion is not merely shifting decimal points; it is the mathematical process of multiplying a quantity by a conversion factor equal to 1. Mastering the behavior of nested prefixes, squared/cubed units, and inverse proportionalities is a core survival skill in advanced physics competitions.

Reference: Essential Conversion Factors

Category	Unit / Quantity	Conversion Equivalent
Length	1 Kilometer (km)	$= 10^3 \text{ m}$
	1 Meter (m)	$= 100 \text{ cm} = 10^3 \text{ mm} = 10^6 \mu\text{m}$
	1 Parsec (pc)	$\approx 3.26 \text{ light-years (ly)}$
Volume	1 Cubic Meter (m^3)	$= 1000 \text{ Liters (L)}$
	1 Liter (L)	$= 10^3 \text{ cm}^3 = 10^{-3} \text{ m}^3$
	1 Fluid Ounce (fl oz)	$\approx 29.57 \text{ cm}^3$
Time	1 Hour (hr)	$= 60 \text{ min} = 3600 \text{ s}$
Mass	1 Kilogram (kg)	$= 10^3 \text{ g}$
	1 Gram (g)	$= 10^3 \text{ mg}$
Pressure	1 Atmosphere (atm)	$= 101,325 \text{ Pascals (Pa)}$

1.1 Multiple Choice Questions

PROBLEM 1

LEVEL I

An engine's temperature increases by 10°C . What is this increase (change) in degrees Fahrenheit ($^{\circ}\text{F}$)?

- (a) 50°F
- (b) 18°F
- (c) -12°F
- (d) 283°F

PROBLEM 2

LEVEL I

A car engine rotates at a speed of $600/\pi$ revolutions per minute (RPM). What is its angular velocity in rad/s ?

- (a) 10 rad/s
- (b) 20 rad/s
- (c) 60 rad/s
- (d) 100 rad/s

PROBLEM 3

LEVEL II

The density of a certain material is 1 mg/mm^3 . What is the value of this density in fundamental SI units (kg/m^3)?

- (a) 1 kg/m^3
- (b) 100 kg/m^3
- (c) $1,000 \text{ kg/m}^3$
- (d) $1,000,000 \text{ kg/m}^3$

PROBLEM 4

LEVEL II

In some countries, fuel efficiency is measured in km/L, while in others it is measured in L/100km. If a car's efficiency is 20 km/L, what is its efficiency in the latter unit?

- (a) 2000 L/100km
- (b) 0.05 L/100km
- (c) 5 L/100km
- (d) 50 L/100km

PROBLEM 5**LEVEL II**

The acceleration due to gravity is approximately 10 m/s^2 . If we want to express it in km/min^2 , we must:

- (a) Multiply by 3.6
- (b) Divide by 1000, then multiply by 60
- (c) Divide by 1000, then multiply by 3600
- (d) Multiply by 1000, then divide by 3600

PROBLEM 6**LEVEL III**

In dynamic fluid mechanics, kinematic viscosity ν is defined as the dynamic viscosity μ ($\text{Pa} \cdot \text{s}$) divided by density ρ (kg/m^3). In the CGS system, kinematic viscosity is measured in *Stokes* ($\text{St} = \text{cm}^2/\text{s}$). If a fluid has a dynamic viscosity of $\mu = 1.2 \text{ cP}$ ($1 \text{ cP} = 10^{-3} \text{ Pa} \cdot \text{s}$) and a density of $\rho = 0.80 \text{ g/cm}^3$, what is its kinematic viscosity in *Stokes*?

- (a) $1.5 \times 10^{-2} \text{ St}$
- (b) $1.5 \times 10^{-1} \text{ St}$
- (c) 1.5 St
- (d) 15 St

1.2 Free-Response Questions

PROBLEM 7**LEVEL I**

In a hospital, a patient is given an intravenous fluid at a flow rate of 2 fluid ounces per hour.

Given that 1 fluid ounce $\approx 29.57 \text{ cm}^3$.

Required: Calculate the flow rate in cubic millimeters per second (mm^3/s).

PROBLEM 8**LEVEL II**

Astronauts discover an alien civilization that uses a completely different system of physical units:

- The unit of mass is the "Zorp" (Z), where $1 \text{ Z} = 1.5 \text{ kg}$.
- The unit of length is the "Blip" (B), where $1 \text{ B} = 0.5 \text{ m}$.
- The unit of time is the "Glop" (G), where $1 \text{ G} = 2 \text{ s}$.

A force of 10 Newtons ($\text{N} = \text{kg} \cdot \text{m}/\text{s}^2$) acts on an object.

Required: Convert this force into the alien units ($\text{Z} \cdot \text{B}/\text{G}^2$), clearly showing your mathematical steps.

PROBLEM 9**LEVEL II**

In the van der Waals equation for real gases, the constant a for oxygen gas has the following value in SI units:

$$a = 0.138 \text{ Pa} \cdot \text{m}^6/\text{mol}^2$$

Given that atmospheric pressure is $1 \text{ atm} = 101,325 \text{ Pa}$ and $1 \text{ L} = 10^{-3} \text{ m}^3$.

Required: Convert this constant to the unit ($\text{atm} \cdot \text{L}^2/\text{mol}^2$).

PROBLEM 10**LEVEL II**

The radius of the observable universe is estimated to be 46.5×10^9 light-years (ly). Given that $1 \text{ Parsec (pc)} = 3.26 \text{ ly}$.

Required: Calculate the volume of the observable universe (treating it as a sphere, $V = \frac{4}{3}\pi r^3$) in cubic parsecs (pc^3).

PROBLEM 11**LEVEL II**

The Stefan-Boltzmann constant is given by:

$$\sigma = 5.67 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4}$$

Given that the Watt $\text{W} = \text{J}/\text{s} = \text{kg} \cdot \text{m}^2/\text{s}^3$.

Required: 1. Rewrite the SI unit of the constant using only fundamental units (without the

Watt). 2. Convert the constant's value to the CGS system, where mass is in grams (g) and length is in centimeters (cm).

PROBLEM 12**LEVEL III**

In natural units (Planck units), fundamental constants such as the speed of light c , the reduced Planck constant $\hbar$, and Newton's gravitational constant G are defined to equal 1. In this system, mass, length, and time can all be expressed in terms of energy (E , measured in electronvolts eV). Given the SI values:

- $c \approx 3.0 \times 10^8 \text{ m/s}$
- $\hbar \approx 1.05 \times 10^{-34} \text{ J} \cdot \text{s}$
- $1 \text{ eV} \approx 1.6 \times 10^{-19} \text{ J}$

Required: Derive the expression for the *Planck length* $l_p = \sqrt{\frac{\hbar G}{c^3}}$ in terms of GeV^{-1} ($1 \text{ GeV} = 10^9 \text{ eV}$), and compute its numerical value given $l_p \approx 1.6 \times 10^{-35} \text{ m}$.

Dimensional Analysis

Topic Note: Dimensional Analysis

Dimensional analysis is a powerful tool in physics used to check the mathematical plausibility of equations, derive empirical formulas, and understand the relationship between different physical quantities. The three fundamental dimensions in mechanics are Mass (M), Length (L), and Time (T).

The Principle of Dimensional Homogeneity: You can only add, subtract, or equate quantities that have the exact same dimensions.

Reference: Common Physical Dimensions

Quantity	Symbol	Dimensional Formula	SI Unit
Mass	m	M	kg
Length	L, x, r, d	L	m
Time	t	T	s
Velocity	v	LT^{-1}	m/s
Acceleration	a, g	LT^{-2}	m/s ²
Force	F	MLT^{-2}	N (Newton)
Energy / Work	E, W, K, U	ML^2T^{-2}	J (Joule)
Power	P	ML^2T^{-3}	W (Watt)
Pressure	P, p	$ML^{-1}T^{-2}$	Pa (Pascal)
Density	ρ	ML^{-3}	kg/m ³
Angle	θ	1 (Dimensionless)	rad

2.1 Multiple Choice Questions

PROBLEM 13

LEVEL I

If the physical equation $X = Y + Z$ is valid, what condition must be met?

- (a) The dimensions of X must equal the dimensions of Y multiplied by Z .
- (b) The dimensions of X , Y , and Z must be exactly identical.
- (c) Y and Z must be dimensionless.

PROBLEM 14

LEVEL I

You are trying to recall the formula for "time" t , where d is distance and v is velocity. Which of the following formulas is **impossible** to be correct?

- (a) $t = \frac{d}{v}$
- (b) $t = \frac{2d}{v}$
- (c) $t = d \cdot v$

PROBLEM 15

LEVEL I

In the wave equation $y = \sin(k \cdot x)$, if x represents "distance", what is the dimension of the constant k ?

- (a) L (Length)
- (b) L^{-1} (Inverse of length)
- (c) Dimensionless

PROBLEM 16

LEVEL I

The area of a circle is $A = \pi \cdot r^2$. What is the dimension of the constant π ?

- (a) Length (L)
- (b) Area (L^2)
- (c) Dimensionless (1)

PROBLEM 17**LEVEL I**

Hooke's Law for a spring states that the force equals the spring constant multiplied by the extension ($F = k \cdot x$). What is the standard unit of the constant k ?

- (a) Newton · meter
- (b) Newton / meter
- (c) Newton

PROBLEM 18**LEVEL I**

A water tank drains at a constant rate R . What are the dimensions of this rate R ?

- (a) L^3T
- (b) L^2T^{-1}
- (c) L^3T^{-1}

PROBLEM 19**LEVEL I**

Mechanical work is defined as (Force $\times$ Distance), and kinetic energy is defined as ($\frac{1}{2} \times$ Mass $\times$ Velocity squared). What is the relationship between their dimensions?

- (a) The dimensions of work are greater.
- (b) They have no relationship.
- (c) They have exactly the same dimensions.

PROBLEM 20**LEVEL I**

Which of the following physical quantities is dimensionless?

- (a) Angle θ
- (b) Acceleration a
- (c) Density ρ

PROBLEM 21**LEVEL I**

If you are told that an unknown physical quantity has the dimensions $(L \cdot T^{-2})$, what is this quantity?

- (a) Velocity
- (b) Acceleration
- (c) Force

PROBLEM 22**LEVEL II**

An object's velocity decreases according to the relation $v = v_0 \cdot e^{-c \cdot t}$, where t is time and e is Euler's number. What is the dimension of the constant c ?

- (a) T (Time)
- (b) T^{-1} (Inverse of time)
- (c) L/T (Velocity)

PROBLEM 23**LEVEL III**

In astrophysics, the Chandrasekhar mass limit M_{Ch} of a white dwarf depends on Planck's constant $\hbar$, the speed of light c , Newton's gravitational constant G , and the mass of a nucleon m_n . Using dimensional analysis, what is the proportional dependence of M_{Ch} on these fundamental constants?

- (a) $M_{\text{Ch}} \propto \left(\frac{\hbar c}{G}\right)^{3/2} m_n^{-2}$
- (b) $M_{\text{Ch}} \propto \left(\frac{\hbar G}{c^3}\right)^{1/2} m_n$
- (c) $M_{\text{Ch}} \propto \frac{\hbar c}{G m_n}$
- (d) $M_{\text{Ch}} \propto \frac{G m_n^2}{\hbar c}$

2.2 Free-Response Questions

PROBLEM 24**LEVEL I**

During a lab experiment, your lab partner proposed the following equation to calculate the

velocity (v) of an object:

$$v = d + a \cdot t$$

(where d is distance, a is acceleration, and t is time).

Explain in a short paragraph, using dimensional analysis, why this equation is mathematically and physically impossible, and state the fundamental principle your partner violated.

PROBLEM 25

LEVEL II

When we use dimensional analysis to derive the kinetic energy formula, we can easily find that energy is proportional to mass and velocity squared ($m \cdot v^2$). However, this method completely fails to tell us about the fraction ($\frac{1}{2}$) in the actual formula ($E = \frac{1}{2}mv^2$).

Write a scientific explanation detailing why dimensional analysis is unable to discover pure mathematical constants and numbers.

PROBLEM 26

LEVEL II

The golden rule states: "If the dimensions of the right side do not equal the dimensions of the left side, the equation is definitely wrong." However, is the reverse true? That is, if you prove that both sides are dimensionally identical, does this necessarily mean the equation is 100% physically correct? Support your answer with a logical argument or example.

PROBLEM 27

LEVEL II

In many physical laws, we use functions like sine (sin) or logarithm (log). Explain why any quantity we place inside these functions (such as the angle θ or a variable exponent) must be a "dimensionless quantity". What would happen physically if we tried to calculate $\sin(5 \text{ meters})$ or $\log(10 \text{ kg})$?

PROBLEM 28

LEVEL III

The speed v of ocean surface waves depends on the wavelength λ , fluid density ρ , gravitational acceleration g , and surface tension γ (where $[\gamma] = MT^{-2}$).

- (a) Using Buckingham's Π theorem, show that the phase velocity of surface waves can be expressed as:

$$v = \sqrt{\frac{g\lambda}{2\pi} + \frac{2\pi\gamma}{\rho\lambda}}$$

- (b) Identify the dominant regime for very long wavelengths ($\lambda \rightarrow \infty$) versus extremely short wavelengths ($\lambda \rightarrow 0$).

PROBLEM 29LEVEL III

When an intense nuclear explosion occurs in air, a spherical blast wave expands outward. The radius $R(t)$ of the shock wave depends on time t , energy released E , and initial air density ρ_0 .

- (a) Use dimensional analysis to derive the relationship between $R(t)$ and time t .
- (b) If the blast wave radius reaches $R_1 = 80$ m at time $t_1 = 1.0$ ms, estimate the radius R_2 at $t_2 = 16.0$ ms.

Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

@hintlist